

Invest **NOVA SCOTIA**

Nova Scotia's Department of Natural Resources and Renewables (NSDNR) and Invest Nova Scotia are partnering to offer the Low Carbon Technology Stream of ESCF and will work together on the evaluation process and curriculum.

Educational Institution: Dalhousie University
Researcher(s): Dr. Peter Bodorik and Dr. Chris Liu, postdoctoral researcher
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I am applying for: Phase One x ☐ Phase Two ☐ • Phase one is for first-time projects with technologies in initial stages of development and an identified opportunity for commercialization. • Phase two is for projects that have identified an opportunity, target markets and potential revenue streams, have initiated discussions with potential customers and have established a collaboration with a commercial partner. Phase two applicants must submit their final phase one report prior to the application deadline.

Project Title: Blockchains: Smart Contract as a Service (SCaaS)		
Lay Description: 50 words or less summary that is non-confidential		
SCaaS helps organizations create blockchain smart contracts while dramatically reducing the costs and need for blockchain experts. It automatically turns business process diagrams into secure, executable smart contracts, making smart contracts easier, cheaper, and safer to use for trade, supply chains, and financial applications.		
Yes	No	
☐	☒	Are you aware of any regulations associated with this technology/product?
☒	☐	Have you filed a patent for this technology?
☐	☒	Have you obtained any customer feedback?
☒	☐	Do you have a prototype?
☐	☒	Have you received other financial assistance for this project? If so, how much in total? \$_____
☒	☐	Do you see this as a licensing opportunity or as a new spin-out? If so, which one? <u>Spin-out</u>
☐	☒	Is this technology assigned to your educational institution?

1.0 ESCF Project Overview (maximum ½ page)

Describe the proposed ESCF project.

Title: *Minimum Viable Product (MVP) for Smart Contract as a Services (SCaaS) from BPMN Models*

The *Smart Contract as a Service (SCaaS)* framework automates the generation of smart contracts from business process models (represented as models using *Business Process Modeling Notation (BPMN)* and *Decision Management Notation (DMN)*) and offers features such as nested multi-step transactions, sidechain cost optimization, and lifecycle support (upgrade/repair). It has been validated as a proof-of-concept in research settings and documented in peer-reviewed publications.

Objective: Prepare the SCaaS prototype as a *commercially viable platform* that enables business analysts to create, deploy, and manage blockchain smart contracts.

Minimum Viable Product (MVP): An MVP will be developed that allows users to (i) model business process visually using BPMN and DMN; (ii) generate and deploy smart contracts automatically from BPMN/DMN inputs; (iii) manage Smart Contract Lifecycle, including versioning, upgrades, and automated repair based on external triggers, and (iv) provide *compliance* through smart contracts in conjunction with off-chain compliance support for search and management of internal and external compliance policies.

This work will involve enabling a *Business Analyst (BA)* to (i) upload BPMN/DMN model as diagrams and configure business rules; (ii) use a cloud backend service to automatically generate a smart contract and deploy it to supported blockchains (initially Ethereum + Hyperledger Fabric); and (iii) deploy and test the smart contract using testnet support, such as Goerli testnet for Ethereum, and sandbox environments for enterprises to trial contract deployments without cost.

2.0 Description of Discovery (maximum 1½ pages)

Describe the background and current stage of the technology, as well as potential applications and markets. What are the major benefits for existing products or solutions? Are there any barriers to adoption? Include any relevant scientific references.

*** Please note references are at the end of this document.**

Background, Technology, and Commercial Opportunity

Blockchain smart contracts promise to automate complex business transactions across organizational boundaries, yet their adoption in real-world trade, supply chain, and financial services remains limited. A major barrier is the difficulty, cost, and risk associated with developing correct and secure smart contracts for complex business workflows. (Zou et al., 2020)

Research conducted at the Dalhousie Blockchain Lab (Dalhousie University, n.d.) addresses this challenge through *Smart Contracts as a Service (SCaaS)*, a model-driven framework that automates the generation of executable smart contracts from standard business process models. Using SCaaS, *Business Analysts (BAs)* define workflows in *Business Process Model and Notation (BPMN)* and decision logic in *Decision Model Notation (DMN)*. These high-level models are automatically transformed into deployable smart contracts and application interfaces, eliminating the need for low-level blockchain programming and reducing dependence on scarce developer expertise.

The SCaaS transformation pipeline converts BPMN models into graph representations and applies a multi-modal modeling approach that integrates *Discrete Event (DE)* and *Hierarchical State Machine (HSM)* semantics. This enables precise modeling of collaboration, temporal dependencies, and functional behavior prior to code generation. As a result, SCaaS supports advanced capabilities that are difficult to implement manually, including nested multi-step transactions, sidechain computation for cost efficiency, smart contract lifecycle management (upgrade and repair), portability across blockchain platforms, and privacy-preserving execution.

The technology has been validated through proof-of-concept implementations, using a research prototype, and multiple peer-reviewed publications demonstrating that automated transformation can generate correct, modular smart contracts that enforce transactional properties across complex workflows. Use cases explored include order processing, collaborative trade, and supply chain coordination, confirming feasibility across diverse domains.

Current State and Need for Funding

SCaaS is currently at a *prototype stage*. The underlying transformation engines, transaction semantics, and lifecycle mechanisms (including earlier versions of the tools, TABS+, TABS+R) have been rigorously evaluated in academic settings. Recent and forthcoming work focuses on usability, lifecycle support, SME accessibility, and payment rails for decentralized finance, signaling readiness to transition toward practical deployment.

However, the current system requires input from early customers for commercialization. Funding is sought to transition SCaaS from an academic prototype to a *minimum viable product (MVP)* that potential customers and investors can evaluate through a “Try SCaaS” option.

Target Applications and Market Opportunity

SCaaS targets sectors where smart contracts can automate complex inter-organizational processes:

- *Trade and Supply Chain Management*: Multi-party workflows involving logistics providers, customs, insurers, banks, and auditors.
- *Finance and Decentralized Finance (DeFi)*: Automated execution of financial agreements with reduced reliance on intermediaries.
- *Enterprise Interoperability*: Workflow automation across heterogeneous systems and organizational boundaries.

Primary market segments include *large enterprises* with complex regulatory requirements, *small and medium enterprises (SMEs)* that lack in-house blockchain expertise, and *consulting and system-integration firms* seeking efficient blockchain deployment tools. Research results indicate that SCaaS is usable by SMEs, significantly expanding its potential market reach.

Value Proposition and Competitive Advantage

SCaaS offers clear advantages over existing smart contract development approaches:

- *Reduced cost and complexity* through automated generation from BPMN/DMN models.
- *Transactional robustness* via support for nested, multi-step transactions and lifecycle management.
- *Platform portability and interoperability* across blockchain technologies.
- *Improved security and privacy* through automated cryptographic and access-control generation.
- *Compliance readiness*, with recent extensions supporting regulatory and legal constraints.

Risks and Mitigation

Key adoption risks include tooling maturity, blockchain platform limitations, and regulatory uncertainty. The proposed funding directly addresses these risks by enabling MVP engineering, usability improvements, maturing compliance-aware features, and real-world pilot deployments.

SCaaS represents a research-validated technology with strong commercial potential. Funding will enable the critical transition from prototype to an MVP platform that lowers barriers to blockchain adoption and unlocks smart contract value for enterprises and SMEs.

3.0 Exposure of the Discovery to Date and Intellectual Property (maximum 1 page)

3.1 Summarize any public exposure your discovery has had to date, such as articles in publications or non-confidential discussions.

The software product is described in peer-reviewed research published in reputable conferences and journals (ACM, Springer, IEEE), which legitimizes SCaaS and encourages adoption by enterprises and research partnerships.

3.2 Confirm who has ownership of the discovery.

Drs. Liu, Bodorik, and Jutla have the ownership of the discovery.

3.3 Describe the intellectual property strategy to protect your discovery. Provide a summary of related patents, prior art and freedom to operate, if applicable.

The project is grounded in peer-reviewed research published in reputable conferences and journals (ACM, Springer, IEEE), which legitimizes SCaaS in a way that many commercial startups cannot easily emulate. This backing also encourages adoption by enterprises and research partnerships, further insulating SCaaS from rapid replication by commercial competitors.

There exist patents that relate *conceptually* to automated generation of smart contract code from high-level inputs that cover some aspects of automation and code generation for multiple blockchains, but none appear to specifically reference BPMN/DMN-to-smart-contract transformation.

Although no patents are currently associated with the SCaaS, we are preparing a submission of USPTO provisional applications on the following recent innovations within the project, including:

- Compliance enforcement for smart contracts
- Repair of smart contracts
- Generation of API for automated generation of smart contracts
- Payment Rails for Smart Contracts

A robust *Freedom-to-Operate (FTO)* analysis should be conducted with professional IP counsel, but preliminary search indicates that the specific “multi-modal modeling of BPMN/DMN for nested transaction generation and automated smart contract code derivation” is distinct from known mainstream patents. This suggests that, without proprietary patent filings yet, the research is likely in a clear space for original IP claims if novel elements can be clearly articulated and distinguished from broader prior art.

4.0 Market Analysis (maximum 2 pages)

4.1 Market assessment: Describe the market potential and target market(s). What is the potential size of the market for your technology? Provide details on the demand for the product and estimated sales (e.g., annual sales of similar products already on the market).

Target Markets – Smart contract automation technologies, like SCaaS, are part of the broader smart contracts market, which includes software tools, platforms, and services for creating, deploying, and managing smart contracts across industries. The *key target market segments are:*

- *Large Enterprises & Corporations:* Financial services, supply chain/logistics, healthcare, real estate, and manufacturing with complex inter-organizational workflows are early adopters of smart contract solutions. These entities benefit from automation, reduced intermediaries, and enhanced transparency.
- *Small and Medium Enterprises (SMEs):* Increasing access to *model-driven, low-code tools* expands adoption beyond enterprises with large blockchain teams.
- *Blockchain Platform Providers & Integrators:* Firms that embed contract automation into broader blockchain ecosystems (e.g., enterprise blockchain integrations, hybrid public/private deployments).

This reflects broader smart contract adoption trends, where large enterprises dominate current usage, but SME adoption is accelerating (Elad, 2025).

Market Size and Forecasts: A variety of industry research forecasts show rapid growth in the smart contracts market, representing a useful proxy for SCaaS demand. Sample forecasts for near, medium, and long-term growth projections for market size are:

- *Near-Term:* The global smart contracts market was valued at about *USD 3.2 billion in 2025*, and is projected to reach *USD 3.94 billion in 2026* at a ~23.1% annual growth rate. (The Business Research Company, 2025)
- *Medium-Term Growth:* Growth to approximately *USD 9.36 billion by 2030* is forecast. (The Business Research Company, 2025)
- *Long-Term Projections:* Other reports estimate the global smart contracts market reaching *USD 24.67 billion by 2034* (Zion Market Research, 2025)

The specific segment of smart contract tools and software, into which SCaaS directly fits, is valued at roughly *USD 1.42 billion in 2024* and is forecast to grow to about *USD 5 billion by 2035* at ~12% CAGR. (WiseGuy Reports, 2025)

Because smart contracts are a core component of most blockchain solutions, the overall blockchain market context is relevant and it has been valued at ~USD 26.9 billion today and is projected to approach nearly *USD 1.88 trillion by 2034*, driven in part by smart contract and decentralized application demand. (GlobeNewswire, 2025)

These figures suggest a multi-billion-dollar addressable market for technologies that simplify smart contract creation, which directly aligns with what SCaaS offers. Furthermore, SCaaS sits at the intersection of these expanding markets, amplifying its potential reach.

4.2 Competitive advantage: What are the technologies that currently serve the target market? List any other products in development expected to provide near-term solutions to similar problems. Explain why your discovery is more effective than alternative technologies.

Technologies Currently Serving the Target Market: In the broader ecosystem today, smart contract capabilities come from (i) general smart contract platforms (e.g., Ethereum, Hyperledger, Solana) that provide execution environments for contracts but still require technical coding. (TechTarget, 2025); (ii) Developer tools (e.g., Solidity compilers, Hardhat, etc.) that help with writing, testing, and auditing smart contracts, but don't automate business modeling into contracts. (Chainlink, n.d.); and (iii) Enterprise Blockchain Frameworks (e.g., Hyperledger Fabric, Corda, Quorum) that support permissioned networks and chaincode, but again require technical developer involvement. (Wikipedia contributors, n.d.)

These technologies are foundational but *do not themselves reduce the need for developer effort in translating business logic into secure contract code.*

Key Advantages of SCaaS approach and tool

a) Lower Cost & Faster Development: Instead of writing code by hand, SCaaS turns high-level business process diagrams directly into smart contracts and APIs, and thus dramatically reducing developer effort, time, and cost.

b) Designed for Business Users: BAs use graphical BPMN and DMN models (standard business modeling languages) to represent both processes and decision logic in a way that's familiar across business and IT teams. This bridges the gap between business intent and technical implementation.

c) Nested Multi-Step Transactions: Blockchain natively supports only single, atomic transactions. SCaaS generates *multi-step, nested transactions* that coordinate complex collaborations (e.g., supply chain with customs, shippers, insurers) with ACID-like guarantees, which a capability that many tools lack.

d) Privacy, Portability & Interoperability: SCaaS logic ensures that participants see only the data relevant to them and can target multiple blockchain platforms (e.g., Ethereum, Hyperledger Fabric). Smart contract methods can interoperate across contracts and chains.

e) Smart Contract Lifecycle Support: Unlike most smart contract tools, SCaaS includes mechanisms for *upgrade* (swap-in new logic), *repair* (continue execution after interruption and repair), and *compliance/reporting*, addressing operational realities and regulatory needs.

4.3 Barrier to competitive entry: What are the sustainable and significant barriers to competitors entering your marketplace?

a) Deep Technical Innovation and IP: The SCaaS approach is built on a *novel multi-modal transformation methodology* that goes beyond simple code templates by providing *nested multi-step transaction support, portability, privacy control, interoperability, and lifecycle mechanisms* such as upgrade and repair that *typical code-generation tools do not offer*.

This technical architecture represents a *sophisticated intellectual property position*: competitors would need to replicate not just GUI tools but also the underlying model-to-contract transformation semantics and multi-chain execution patterns. This depth of modeling and automation is *not a trivial engineering problem*, and it's supported by multiple peer-reviewed publications and tools (e.g., (Liu, Bodorik & Jutla, 2024; Liu, Bodorik & Jutla, 2025)).

b) Standards & Model-Driven Engineering Expertise: SCaaS leverages *industry standards (BPMN, DMN)* to let non-programmers define complex distributed systems and then auto-generate secure blockchain logic. Competitors must match this blending of *business modeling standards with blockchain execution platforms*. Without deep expertise in both *formal modeling and distributed ledger internals*, new entrants will struggle to produce tools that are equally accessible to business analysts and robust enough for real applications.

c) Lifecycle & Repair Capabilities: Most blockchain tools focus narrowly on contract deployment, with limited support for *post-deployment lifecycle management*. SCaaS explicitly integrates upgradability and programmatic repair, thus enabling continuity of business logic when external conditions change (e.g., logistical delays, regulatory updates).

d) *Interoperability and Cross-Chain Support*: SCaaS can generate logic for multiple platforms (e.g., Ethereum, Hyperledger Fabric) and model interoperability across them by design. This is a *non-trivial engineering and maintenance problem*.

e) *Research Backing and Academic Credibility*: The project is grounded in peer-reviewed research published in reputable conferences and journals (ACM, Springer, IEEE), which legitimizes SCaaS in a way that many commercial startups cannot easily emulate. This backing also encourages adoption by enterprises and research partnerships, further insulating SCaaS from rapid replication by commercial competitors.

f) *First-Mover Advantage in Model-Driven Smart Contracts*: Although there are emerging frameworks for smart contract automation and *blockchain services as a service* in the broader ecosystem, few (if any) fully combine *business analyst reuse, nested transaction enforcement, lifecycle support, and sidechain optimization* in a single, integrated framework. SCaaS therefore enjoys a *first-mover advantage in a specific niche* of business-model-driven blockchain solutions, creating inertia and early adoption benefits over latecomers.

5.0 LOW CARBON STREAM APPLICANTS ONLY (maximum 1 page) - N/A

5.1 Carbon intensity reduction: What is the estimated impact on carbon emissions reduction (megatonnes of CO₂ equivalent per year) if your technology is adopted by 100 per cent of the market for the application you are targeting? What are the technologies that currently serve the target market? Explain why your discovery will be more effective than alternative technologies.

5.2 Performance metrics: Identify the key performance metrics that will be assessed in the project. Describe what these metrics need to achieve to justify further development work and future commercialization.

6.0 Development Plan (maximum 2 pages)

6.1 Describe the steps in technology development over the term of the proposed project. Proposed activities may include: intellectual property development; development of intellectual property protection strategies; assessment of market potential; bench-stage analysis of technology performance; development of go-to-market strategies, business plans and strategic plans; proof-of-concept or prototype development; technology design and optimization; pre-clinical testing; clinical validation; and leveraging other funds.

Over the term of the proposed project, we will focus on transforming the existing SCaaS research proof of concept into a usable, market-ready prototype that can be evaluated by potential customers, partners, and investors.

The first step will be technology stabilization and refinement. Although the core transformation engines have been validated in research, they need engineering work to improve robustness, error handling, and performance. This includes hardening the BPMN/DMN-to-smart-contract transformation pipeline, improving transaction handling, and ensuring reliable deployment to supported blockchain platforms.

In parallel, we will develop an intellectual property and commercialization strategy. This includes reviewing existing publications, identifying protectable innovations (e.g., lifecycle management, repair mechanisms, payment rails), and preparing provisional patent filings

where appropriate. Market feedback gathered during the project will inform which components are prioritized for protection.

The next step is MVP development. We will build a usable SCaaS MVP that allows business analysts to upload BPMN/DMN models, configure options, and automatically generate and deploy smart contracts in a controlled test environment. This includes developing a basic user interface, cloud backend services, and testnet deployment support.

We will then conduct market validation and customer discovery activities. This includes engaging with potential enterprise users, SMEs, and consulting partners to gather feedback on usability, value proposition, pricing expectations, and integration needs. This feedback will guide refinements to the prototype and inform go-to-market planning.

Finally, we will prepare commercialization readiness materials, including a technical roadmap, early business model options (licensing vs. spin-out), and identification of strategic partners. These outputs will position SCaaS for follow-on funding, pilot projects, or spin-out formation after the ESCF period.

- 6.2 Provide detailed timelines and deliverables for the term of the proposed project. Identify the critical stages in the development plan and required steps after the funding period to successfully commercialize the technology.

The proposed project will be completed over a 12-month period and is organized into four main stages.

Stage 1 (Months 1–3) – Technology Refinement and Planning: Key activities include system architecture for the commercial MVP and initiating intellectual property review.

Deliverables: Refined technical architecture, prioritized feature list, IP strategy outline.

Stage 2 (Months 4–6): Prototype Engineering: During this stage, we will implement the SCaaS MVP, including BPMN/DMN ingestion, automated smart contract generation, deployment to supported blockchains, and basic user interaction.

Deliverables: Working SCaaS prototype with testnet deployment capability.

Stage 3 (Months 7–9) – Validation and Market Feedback: The MVP will be shared with selected users and partners for hands-on evaluation. Feedback will be collected on usability, functionality, and commercial relevance.

Deliverables: User feedback reports, refined MVP, preliminary go-to-market insights.

Stage 4 (Months 10–12) – Commercialization Preparation: The final stage focuses on preparing for post-funding commercialization. This includes refining the business model, identifying pilot opportunities, and preparing materials for investors or partners.

Deliverables: Commercialization roadmap, partner shortlist, next-phase funding and deployment plan.

Post-Funding Steps: After the ESCF funding period, the next critical steps include launching pilot deployments with early adopters, securing strategic or investment partners, and progressing toward a formal spin-out.

- 6.3 Outline a potential exit strategy (e.g., spin-out or licensing). What key research, regulatory, and strategic partnership issues need to be addressed to move the discovery to market? Who are the potential regional, national or multinational partners for the technology?

Potential exit strategy: SCaaS is well suited for a university spin-out focused on delivering *Smart Contracts as a Service* as a commercial SaaS platform. A spin-out enables tighter control over product direction, customer engagement, and IP development, while capturing long-term value through subscription licensing, enterprise contracts, and professional services. This path supports iterative product refinement and pilot deployments with early adopters in trade, supply chain, and financial services. Once the initial spin-out matures the technology and market traction, this can be followed up by acquisition or strategic licensing.

Regulatory and Legal Considerations: Smart contract legal enforceability and compliance vary across jurisdictions and industries, while alignment with regulatory requirements in finance, trade, data protection (e.g., privacy and auditability), and cross-border transactions is essential. *SCaaS supports compliance through supporting services* that allow legal and compliance experts to specify the compliance requirements for various jurisdictions and products, while generated smart contracts provide for automated invocation of services to check (i) which regulations are applicable to a smart contract and (ii) whether the smart contract complies with the applicable regulations.

- 6.4 If applicable, outline any regulatory steps or key milestones relating to approval of the technology in Canada or the United States (or other jurisdiction). Also describe any steps undertaken towards regulatory approval.

7.0 Project Team (maximum 1½ pages)

- 7.1 Describe the expertise and contribution of each project team member and how it relates to the accomplishment of project milestones, both in product development and commercialization. Include industry collaborators, if applicable.

The SCaaS project is led by a multidisciplinary team whose combined expertise spans model-driven software engineering, business process modeling, blockchain systems, and technology commercialization. The team's complementary skills ensure coverage of the full innovation lifecycle, from foundational research through prototype development and market entry.

Dr. Christian (Gang) Liu, a graduate of Dalhousie University, brings deep expertise in model-driven engineering, software architecture, and automated system transformation. His work focuses on converting high-level business and system models into executable software artifacts with strong guarantees of correctness, modularity, and scalability.

Contribution to Project Milestones:

- Leads the engineering and refinement of the SCaaS transformation pipeline, including BPMN/DMN-to-graph and multi-modal (DE/HSM) model transformations.
- Oversees the implementation of automated smart contract generation, ensuring correctness, transactional integrity, and robustness.
- Drives the transition from research prototypes to production-quality components, supporting milestones related to prototype stabilization, performance tuning, and MVP readiness.

Dr. Liu's contributions are essential to achieving technical maturity and reliability, which are prerequisites for commercial deployment and customer trust.

Dr. Peter Bodorik contributes extensive expertise in business process modeling, decision support systems, and enterprise information systems, with a strong focus on aligning technical solutions with real organizational practices and managerial needs.

Contribution to Project Milestones:

- Leads the business modeling and decision-logic aspects of SCaaS, ensuring BPMN and DMN representations accurately capture real enterprise workflows.
- Guides usability and user-experience design for Business Analysts, particularly within SMEs, lowering adoption barriers.
- Supports market validation through realistic industry use cases, helping prioritize features that deliver measurable business value.

Dr. Bodorik's role is central to commercialization milestones related to user adoption, customer fit, and demonstrable return on investment.

Dr. Dawn Jutla brings substantial experience in distributed systems, blockchain technologies, electronic commerce, and industry-engaged research, with a strong record of translating academic innovation into practical, deployable solutions.

Contribution to Project Milestones:

- Leads blockchain platform strategy and integration, including deployment across Ethereum- and Hyperledger-based environments.
- Oversees security, privacy, and compliance considerations, ensuring the generated smart contracts meet enterprise and regulatory requirements.
- Drives the commercialization and partnership strategy, including industry pilots, engagement with system integrators, and interaction with potential customers and investors.

Dr. Jutla's expertise directly supports milestones related to pilot deployment, regulatory readiness, and go-to-market execution.

Together, the team forms a cohesive execution unit in which Dr. Liu ensures technical correctness and scalability; Dr. Bodorik ensures usability and business relevance, and Dr. Jutla ensures deployability, compliance, and commercialization readiness. This integrated expertise positions the project to deliver a commercial-ready SCaaS prototype, validate it through real-world pilots, and advance toward successful commercialization via spin-out.

8.0 Budget

8.1 Provide a preliminary budget, including cost of research, consulting fees, patenting expenses, etc. List any potential contributions from strategic partners or industrial alliances. Patent costs are limited to 10 per cent of the funding request. Note: Overhead and administration costs are not eligible.

Preliminary Budget:

100% of the budget is dedicated for remuneration of Dr. Liu. A prerequisite for the commercialization is the engineering development of the MVP. The *obvious and necessary person* for this task is Dr. Liu who developed the software. His work will be informed by the other team members' commercialization efforts via customer interaction with the MVP.

8.2 Include all project funding from other sources (e.g., Springboard, ACOA, NSERC, industry partners, etc.) and indicate the percentage of overlap between funding sources. Also indicate sources of funding for other projects and the percentage of overlap with the ESCF project for which you are applying.

None.

Applicant Authorization

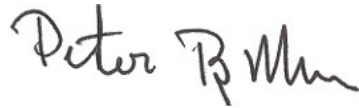
I understand I am obligated to comply with the Early Stage Commercialization Fund funding conditions and reporting requirements [outlined in the ESCF Guidelines](#) if this application is approved.

Peter Bodorik

Professor

Name

Title



February 7

Signature

Date

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